1. CREATING A SIMPLE LAN NETWORK USING CISCO PACKET TRACER

AIM:

To create a simple LAN with 2 PCs and 1 switch and verify connectivity using ping.

STEPS:

- Open Cisco Packet Tracer. From End Devices drag 2 PCs; from Switches drag 1 switch (2960).
- Connections (lightning icon) -> Copper Straight-Through: PC0 Fa0 -> Switch Fa0/1, PC1 Fa0 -> Switch Fa0/2.
- PC0 -> Desktop -> IP Configuration: IP 192.168.1.1, mask 255.255.255.0.
- PC1 -> Desktop -> IP Configuration: IP 192.168.1.2, mask 255.255.255.0.
- PC0 -> Desktop -> Command Prompt and type the ping command below.

COMMANDS:

Command:

```
ping 192.168.1.2
```

OUTPUT:

```
C:\> ping 192.168.1.2
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:

A simple LAN was created and the two PCs communicated through the switch.

2. TO FIND NETWORK IP ADDRESS OF A MACHINE, TIME AND DATE SERVER (1A)

AIM:

To find the IP address of a machine and get the time and date from a server using NTP.

STEPS:

- Place a PC, Server, Switch and Router. Connect with copper straight-through cables (PC-Switch, Server-Switch, Switch-Router).
- PC0: IP 192.168.1.10, mask 255.255.255.0, gateway 192.168.1.1.
- Server: IP 192.168.1.100. Server -> Services -> turn NTP ON and set the time (FTP/HTTP optional).
- Router Gi0/0: 192.168.1.1, then no shutdown.
- On the router, point to the NTP server and check the time. On the PC, type ipconfig to see its IP.

COMMANDS:

Router commands:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1
   255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ntp server 192.168.1.100
Router(config)# end
Router# show ntp status
Router# show clock
```

OUTPUT:

```
Router# show ntp status
Clock is synchronized, stratum 2, reference is
   192.168.1.100
nominal freq is 250.0000 Hz, actual freq is
   249.9990 Hz
reference time is EBDB5DFB.0000028A (20:50:3.650
   UTC Tue Jun 24 2025)
Router# show clock
20:50:34.362 UTC Tue Jun 24 2025
C:\> ipconfig
FastEthernet0 Connection:
   IPv4 Address.......: 192.168.1.10
   Subnet Mask........: 255.255.255.0
   Default Gateway....: 192.168.1.1
```

RESULT:

The IP address of the PC and the time and date of the server were obtained successfully.

3. WORKING WITH COMPLEX (HYBRID) TOPOLOGY (1B)

AIM:

To build a hybrid topology (star + bus) with two routers, two switches and six PCs, and test end-to-end connectivity.

STEPS:

- Devices: 2 routers (backbone), 2 switches (star segments), 6 PCs (3 per switch).
- LAN 1: 192.168.1.0/24 - PC0 .2, PC1 .3, PC2 .4, gateway 192.168.1.1.
- LAN 2: 192.168.2.0/24 - PC3 .2, PC4 .3, PC5 .4, gateway 192.168.2.1.
- Router link: 10.0.0.0/30 (Router0 = 10.0.0.1, Router1 = 10.0.0.2).
- Configure each router in the CLI, add a static route to the other LAN, then ping PC0 -> PC5 (192.168.2.4).

COMMANDS:

Router0:

```
enable
configure terminal
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
interface GigabitEthernet0/0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router1:

```
enable
configure terminal
interface GigabitEthernet0/0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
interface GigabitEthernet0/0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

OUTPUT:

```
C:\> ping 192.168.2.4
Pinging 192.168.2.4 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.4: bytes=32 time=1ms TTL=126
Reply from 192.168.2.4: bytes=32 time=1ms TTL=126
Reply from 192.168.2.4: bytes=32 time<1ms TTL=126
Packets: Sent = 4, Received = 3, Lost = 1 (25%
   loss)
```

RESULT:

The hybrid topology was created and PCs in different LANs communicated through the routers.

4. CONSTRUCTION OF ETHERNET CONNECTION (2C)

AIM:

To construct an Ethernet connection between a PC and a Cisco router in Packet Tracer.

STEPS:

- Connect PC FastEthernet0 to Router Gi0/0 (copper cross-over; Packet Tracer may accept straight-through).
- PC: IP 192.168.1.2, mask 255.255.255.0, gateway 192.168.1.1.
- Configure the router interface and bring it up.
- Ping the router IP from the PC.

COMMANDS:

Router commands:

```
Router> enable
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.1.1
   255.255.255.0
Router(config-if)# no shutdown
```

OUTPUT:

```
C:\> ping 192.168.1.1
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:

The Ethernet connection was established and verified using ping.

5. VERIFY AND CONFIGURE VLAN AND VLAN TRUNK IN PACKET TRACER (2D)

AIM:

To configure and verify VLANs and a trunk link in Cisco Packet Tracer.

STEPS:

- Create VLAN 10 (Faculty) and VLAN 20 (Students) on each switch.
- Assign access ports: fa0/2-4 to VLAN 10, fa0/5 to VLAN 20.
- Configure the link between the switches (g0/1) as a trunk on both ends.
- Verify with show vlan brief and show interfaces trunk.
- Same VLAN: ping works. Different VLANs: need a router (router-on-a-stick with sub-interfaces).

COMMANDS:

Switch commands (on each switch):

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Faculty
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Students
Switch(config-vlan)# exit
Switch(config)# interface range fa0/2-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface fa0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

Trunk (both ends):

```
Switch(config)# interface g0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan
   10,20
Switch(config-if)# exit
Switch# copy running-config startup-config
```

OUTPUT:

```
Switch# show vlan brief
VLAN Name      Status  Ports
10   Faculty   active  Fa0/2, Fa0/3, Fa0/4
20   Students  active  Fa0/5
Switch# show interfaces trunk
Port    Mode  Encapsulation  Status    Native vlan
Gig0/1  on    802.1q         trunking  1
Port    Vlans allowed on trunk
Gig0/1  10,20
```

RESULT:

VLANs and the VLAN trunk were configured and verified.

6. VPN / IPSEC CONFIGURATION (2B SITE-TO-SITE VPN)

AIM:

To configure a site-to-site IPSec VPN between two branch offices over an untrusted network.

STEPS:

- Topology: PC1 - Router1 - ISP Router - Router2 - PC2. Branch A 192.168.1.0/24, Branch B 192.168.2.0/24.
- WAN: Router1 S0/0/0 10.0.0.1 <-> ISP S0/0/0 10.0.0.2 ; ISP S0/0/1 10.0.0.5 <-> Router2 S0/0/0 10.0.0.6 (/30).
- PC1 192.168.1.10 (gw .1), PC2 192.168.2.10 (gw .1). Configure IPs and static routes. clock rate 64000 on the DCE side only.
- Phase 1 (ISAKMP): policy with encryption, hash, authentication, DH group, lifetime; then the preshared key (must match on both routers).
- Phase 2 (IPSec): transform-set, crypto map, ACL 101 for interesting traffic. Apply the crypto map on the outgoing serial interface.
- Verify with show crypto isakmp sa / show crypto ipsec sa and ping PC1 to PC2.

COMMANDS:

Router1:

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
interface Serial0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
ip route 192.168.2.0 255.255.255.0 10.0.0.2
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
lifetime 86400
exit
crypto isakmp key cisco123 address 10.0.0.6
crypto ipsec transform-set MYSET esp-aes esp-sha-
   hmac
crypto map MYMAP 10 ipsec-isakmp
set peer 10.0.0.6
set transform-set MYSET
match address 101
exit
access-list 101 permit ip 192.168.1.0 0.0.0.255
   192.168.2.0 0.0.0.255
interface Serial0/0/0
crypto map MYMAP
```

ISP Router:

```
interface Serial0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
interface Serial0/0/1
ip address 10.0.0.5 255.255.255.252
clock rate 64000
no shutdown
```

Router2:

```
Same as Router1 with: FastEthernet0/0 192.168.2.1,
    Serial0/0/0 10.0.0.6/30,
ip route 192.168.1.0 255.255.255.0 10.0.0.5,
   crypto isakmp key cisco123 address 10.0.0.1,
set peer 10.0.0.1, and ACL 101 reversed:
access-list 101 permit ip 192.168.2.0 0.0.0.255
   192.168.1.0 0.0.0.255
```

OUTPUT:

```
Router1# show crypto isakmp sa
dst         src         state    conn-id slot status
10.0.0.6    10.0.0.1    QM_IDLE  1001    0    ACTIVE
C:\> ping 192.168.2.10
Reply from 192.168.2.10: bytes=32 time=2ms TTL=125
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:

A site-to-site VPN was configured and secure communication between the two branches was verified.

7. CONFIGURE A NETWORK TOPOLOGY USING PACKET TRACER SOFTWARE (3A)

AIM:

To configure a basic network topology with 2 PCs, 2 switches and 1 router using static IP addressing.

STEPS:

- Topology: PC0 --- SW0 --- R0 --- SW1 --- PC1.
- Place 2 PCs, 2 switches (2960) and 1 router. Connect: PC0-Switch0 (Fa0), PC1-Switch1 (Fa0), Switch0-Router0 (Gi0/0), Router0 (Gi0/1)-Switch1. Wait for green links.
- PC0: 192.168.1.10 / 255.255.255.0 / gateway 192.168.1.1. PC1: 192.168.2.10 / 255.255.255.0 / gateway 192.168.2.1.
- Configure both router interfaces (below), then ping 192.168.2.10 from PC0.

COMMANDS:

Router0:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1
   255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1
   255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write
```

OUTPUT:

```
C:\> ping 192.168.2.10
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:

The basic topology was configured and PC0 reached PC1 through the router.

8. TO CONFIGURE SIMPLE STATIC ROUTING (3B)

AIM:

To configure static routes on two routers so PCs in two different networks can communicate.

STEPS:

- Topology: PC0 - Switch0 - Router0 - Router1 - Switch1 - PC1. Network A 192.168.1.0/24, router link 10.0.0.0/30, Network B 192.168.2.0/24.
- Connect PC-Switch and Switch-Router (G0/0) with straight-through; Router0 G0/1 <-> Router1 G0/1 with copper cross-over.
- PC0: 192.168.1.10 (gw 192.168.1.1). PC1: 192.168.2.10 (gw 192.168.2.1).
- Set router interface IPs, no shutdown, add static routes: ip route <destination> <mask> <next-hop>.
- Verify with show ip route (code S) and ping both ways. Static routes are manual, have no overhead, but do not adapt to failures.

COMMANDS:

Router0:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1
   255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1
   255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ip route 192.168.2.0 255.255.255.0
    10.0.0.2
Router(config)# exit
```

Router1:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.2.1
   255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.2
   255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# ip route 192.168.1.0 255.255.255.0
    10.0.0.1
Router(config)# exit
```

OUTPUT:

```
Router0# show ip route
C  10.0.0.0/30 is directly connected,
   GigabitEthernet0/1
C  192.168.1.0/24 is directly connected,
   GigabitEthernet0/0
S  192.168.2.0/24 [1/0] via 10.0.0.2
C:\> ping 192.168.2.10
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:
Static routing was configured and the two networks communicated.

9. TO CONFIGURE NETWORK STATE (LINK-STATE) ROUTING PROTOCOL - OSPF (3D)

AIM:
To configure OSPF, a link-state routing protocol, so routers learn routes automatically.

STEPS:

- Topology: PC0 --- R1 --- R2 --- PC1. Network A 192.168.1.0/24, link 10.0.0.0/30, Network C 192.168.2.0/24.
- Connect PC0 -> R1 G0/0, PC1 -> R2 G0/0 and R1 G0/1 <-> R2 G0/1 (use copper cross-over for router-to-router and PC-to-router; optional switches for realism).
- PC0: 192.168.1.10 (gw 192.168.1.1). PC1: 192.168.2.10 (gw 192.168.2.1).
- Each router floods link-state info, builds a map and runs Dijkstra's shortest-path algorithm. OSPF uses a wildcard mask (inverse of subnet mask: 255.255.255.0 -> 0.0.0.255).
- Verify with show ip ospf neighbor, show ip route (code O), then ping PC0 -> PC1.

Device	Interface	IP Address	Role
PC0	NIC	192.168.1.10	End host
Router1	G0/0	192.168.1.1	LAN side
Router1	G0/1	10.0.0.1	OSPF link
Router2	G0/1	10.0.0.2	OSPF link
Router2	G0/0	192.168.2.1	LAN side
PC1	NIC	192.168.2.10	End host

COMMANDS:
Router1:

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit
```

Router2:

```
enable
configure terminal
interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
interface g0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
exit
```

OUTPUT:

```
Router1# show ip ospf neighbor
Neighbor ID  Pri  State     Dead Time  Address
   Interface
10.0.0.2     1    FULL/DR   00:00:33   10.0.0.2
   Gi0/1
Router1# show ip route
O  192.168.2.0/24 [110/2] via 10.0.0.2,
   GigabitEthernet0/1
C:\> ping 192.168.2.10
Packets: Sent = 4, Received = 4, Lost = 0 (0%
   loss)
```

RESULT:
OSPF was configured and the routers learned routes dynamically.

10. STUDY OF NETWORK SIMULATOR (NS) AND SIMULATION OF CONGESTION CONTROL ALGORITHMS USING NS (3E)

AIM:
To study the Network Simulator (NS2) and simulate a congestion control algorithm (TCP) using NS.

STEPS:

- NS overview: NS2 is an open-source, discrete-event, CLI-driven network simulator (periodic releases, e.g. ns-2.26; runs on FreeBSD, Linux, Solaris, Windows, Mac). Other simulators: OPNET, NetSim (proprietary).
- Functionalities: routing, transport, traffic sources, queuing disciplines, QoS; wireless (ad hoc routing, mobile IP, sensor-MAC); tracing, visualization and utilities.
- The model describes nodes, links and events (transmissions, errors). The main output is the trace file, which logs every packet/event for analysis; NAM gives visual animation. Events are kept in a list and processed in order, one event triggering future events.
- Uses: faster and cheaper than a physical test bed; test new protocols in a controlled, reproducible environment; simulate WAN (TCP, ATM, IP) and LAN (Ethernet, token ring) technologies.
- Performance measures - Packet loss: packets that fail to reach the destination (e.g. queue overflow); high loss makes it hard for transport protocols to keep high bandwidth. Throughput: average rate of successful delivery, in bits/s (or packets/s). Delay: time for a packet to travel from source to destination; end-to-end delay is calculated.
- Simulation procedure: write a Tcl script (nodes, links, TCP agent + FTP source, sink), run with ns, view in NAM, analyse out.tr for throughput, loss and delay.

COMMANDS:
Sample Tcl script (TCP congestion control, 3 nodes with a bottleneck link):

```
set ns [new Simulator]
set nf [open out.nam w]
$ns namtrace-all $nf
set tr [open out.tr w]
$ns trace-all $tr
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
$ns duplex-link $n0 $n1 10Mb 10ms DropTail
$ns duplex-link $n1 $n2 1Mb 20ms DropTail   ;#
   bottleneck
$ns queue-limit $n1 $n2 10
set tcp [new Agent/TCP/Reno]     ;# try Agent/TCP
   (Tahoe), /Newreno, /Vegas
set sink [new Agent/TCPSink]
$ns attach-agent $n0 $tcp
$ns attach-agent $n2 $sink
$ns connect $tcp $sink
set ftp [new Application/FTP]
$ftp attach-agent $tcp
proc finish {} {
  global ns nf tr
  $ns flush-trace
  close $nf
  close $tr
  exec nam out.nam &
  exit 0
}
$ns at 0.5 "$ftp start"
$ns at 10.0 "$ftp stop"
$ns at 10.5 "finish"
$ns run
```

Run:

```
ns tcp.tcl
```

OUTPUT:

```
Trace file lines (out.tr):
+ 0.5    0 1 tcp 1040 ------- 1 0.0 2.0 0 0
- 0.5    0 1 tcp 1040 ------- 1 0.0 2.0 0 0
r 0.5108 0 1 tcp 1040 ------- 1 0.0 2.0 0 0
d 3.21   1 2 tcp 1040 ------- 1 0.0 2.0 57 211
   <- packet dropped (congestion)
(+ enqueue, - dequeue, r receive, d drop)
```

RESULT:
NS2 was studied and a TCP congestion control simulation was run; throughput, packet loss and delay were analysed from the trace file.

11. CASE STUDY - STUDY OF DIFFERENT TYPES OF NETWORK CABLES; CROSS-WIRED AND STRAIGHT CABLE USING CLAMP (CRIMPING) TOOL

AIM:
To study the different types of network cables and practically make a straight-through and a crossover cable using a crimping tool.

STEPS:

- Twisted pair: UTP (Cat 5e, Cat 6; no shielding) and STP (foil/braid shielding for less EMI). Max ~100 m; Cat 5e up to 1 Gbps.
- Coaxial: central copper conductor, insulation, braided shield, jacket. Used in early Ethernet (10BASE2/10BASE5), now cable TV/broadband; 200-500 m segments; good EMI protection.
- Fibre optic: glass/plastic core + cladding, light by total internal reflection. Single-mode (long distance, km to tens of km) and Multi-mode (shorter, building level). Multi-Gbps, immune to EMI, highly secure.
- Straight-through: same standard at both ends (T568B-T568B). Connects different devices: PC-Switch, PC-Router. Crossover: T568A at one end, T568B at the other. Connects similar devices: PC-PC, Switch-Switch (modern ports often have auto MDI-X).

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

COMMANDS:
Algorithm (crimping):

```
1. Cut cable to length; strip ~25 mm of jacket
   without nicking the wires.
2. Untwist the 4 pairs, straighten the 8 wires,
   arrange in T568B/T568A order.
3. Trim the ends flat so ~12 mm extends beyond the
    jacket.
4. Insert into RJ45 (clip down, gold pins facing
   you); jacket seats inside for strain relief.
5. Crimp with the tool until it ratchets/locks;
   press again if needed.
6. Inspect, then test with a cable tester.
7. Repeat on the other end: same standard =
   straight, opposite standard = crossover.
Crossover swap (10/100 Mbps): pins 1<->3 and 2<->6
    (green <-> orange pairs).
```

RESULT:
Straight-through and crossover cables were made using the crimping tool and verified with a cable tester (straight: same pin numbers; crossover: swapped pins).

12. CASE 2 - STUDY OF NETWORK DEVICES IN DETAIL

AIM:
To study the functions, roles and interconnections of key networking devices (repeater, hub, switch, bridge, router, gateway, etc.) in order to build a simple LAN.

STEPS:

- Network devices enable communication, data transfer and network management. Their roles are best understood by OSI layer, as tabulated below.

Device	Function	OSI Layer	Use case
Hub	Broadcasts incoming data to all ports	1 - Physical	Simple networks; obsolete
Repeater	Amplifies/regenerates signals	1 - Physical	Extending range
Bridge	Filters traffic between 2 segments	2 - Data Link	Segmenting networks
Switch	Forwards by MAC address within LAN	2 (some 3)	Efficient LAN transfer
Router	Routes between networks using IP	3 - Network	LAN to internet
Gateway	Translates between different protocols	All, mainly 3	Connecting dissimilar networks
Access Point	Wireless connectivity for a wired LAN	2 - Data Link	Wi-Fi access
Firewall	Filters traffic by security rules	3 and 4	Protection from threats
Brouter	Bridge + router functions	2 and 3	Mixed network types
NIC	Physical interface of a computer	2 - Data Link	Network connectivity

RESULT:
The functions and OSI layers of the major network devices were studied.

13. ROUTER AND SWITCH SECURITY CONFIGURATION (3C)

AIM:
To implement basic security on interconnecting devices (router R1 and switch SW1) using Cisco Packet Tracer.

STEPS:

- Routers and switches are the most vulnerable points in a network, so secure them with: enable secret, console and VTY passwords, password encryption, login banner, and shutting unused switch ports.
- Router: open CLI (press Enter) and apply the router commands. Switch: same steps, then disable unused ports (assume Fa0/10 to 0/24 unused).
- Verify with show running-config, show line and show interfaces status.

COMMANDS:
Router R1:

```
Router> enable
Router# configure terminal
Router(config)# enable secret cisco123
Router(config)# line console 0
Router(config-line)# password conpass
Router(config-line)# login
Router(config-line)# exit
Router(config)# line vty 0 4
Router(config-line)# password vtypass
Router(config-line)# login
Router(config-line)# exit
Router(config)# service password-encryption
Router(config)# banner motd #Unauthorized Access
   is Prohibited!#
```

Switch SW1:

```
Switch> enable
Switch# configure terminal
Switch(config)# enable secret cisco123
Switch(config)# line console 0
Switch(config-line)# password conpass
Switch(config-line)# login
Switch(config-line)# exit
Switch(config)# line vty 0 4
Switch(config-line)# password vtypass
Switch(config-line)# login
Switch(config-line)# exit
Switch(config)# service password-encryption
Switch(config)# banner motd #Access restricted to
   authorized personnel only#
Switch(config)# interface range fastEthernet 0/10
   - 24
Switch(config-if-range)# shutdown
Switch(config-if-range)# exit
```

OUTPUT:

```
Router# show running-config
enable secret 5 $1$mERr$...            <-
   encrypted
line con 0
 password 7 08314D5D1A0E...           <- encrypted
    by service password-encryption
banner motd ^CUnauthorized Access is Prohibited!^C
Switch# show interfaces status
Fa0/10 ... disabled   (ports Fa0/10 - Fa0/24 shut
   down)
```

RESULT:
Basic security was configured on the router and switch and verified.